Ofqual QAN	**600/0406/X, 600/0406/X**
Quartz ID	**1053, 1157**
Qualification	**Level 4 Diploma in Computing, Level 4 Diploma in Computing with Business Management**

Assessment	**Object-Oriented System Analysis and Design**
Assessment Type	**Examination**
Date	**Tuesday 4th March 2025**
Time	**09.00am – 12.00pm**

Examination Paper

Answer ALL questions.

Clearly cross our surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

a) Define the following terms in the context of Object-Oriented Analysis and Design:

i) Coupling **1**

ii) Abstraction **2**

iii) Refactoring **2**

b) Multiple choice questions: Please select ONE (1) from the options provided. Write your answer in the answer booklet.

i) The Waterfall model can be used in object-oriented development when: **1**

a. a working program is required quickly
b. the project has a large development team
c. requirements are well defined
d. legacy systems are developed

ii) Unit testing evaluates **1**

a. glass-box testing
b. execution of individual methods
c. integration of subroutines
d. groups of classes that collaborate or communicate in some way

iii) Object Constraint Language (OCL) can be used for: **1**

a. coding embedded systems
b. unit Testing
c. ensuring cohesion
d. describing pre- and post-conditions on operations and methods

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iv) In object-oriented design, methods and attributes from a superclass being available to a subclass is called: **1**

a. cohesion
b. decoupling
c. inheritance
d. prioritisation

v) In object-oriented design, sufficiency and cohesion can be measured by: **1**

a. number of subclasses
b. number of responsibilities
c. number of attributes
d. all of the above

Total 10 marks

Marks

Question 2

Identify **and** discuss **FIVE (5) disadvantages** of the Agile Methodology. **10**

Total 10 marks

Marks

Question 3

a) Explain what is meant by Use-Case Points (UCP) estimation. **4**

b) Explain **THREE** (3) merits of UCP. **3**

c) Explain **THREE** (3) challenges of using UCP. **3**

Total 10 marks

Marks

Question 4

Compare and contrast object-oriented programming (OOP) with object-based programming (OBP)?

Total 10 marks

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Marks

Question 5

Compare and contrast State Diagrams **and** Activity Diagrams. **10**

Total 10 marks

Marks

Question 6

Demonstrate your understanding of CRC (Class-Responsibility-Collaborator) cards by creating a CRC card for a Patient class for a Hospital System. **10**

Show at least ONE (1) superclass and at least FOUR (4) subclasses that might be associated with a Patient class in a typical Hospital System.

Make any assumptions you believe applicable regarding the responsibilities, collaborators and attributes of the given Patient class.

Total 10 marks

Marks

Question 7

You have been asked to develop a crop management system for a farmer. The farmer wishes to know which crops are grown in which field, when the crops are planted, harvested and the yield. **10**

The system also needs to keep track of the maintenance of the crops. All records need to include the name of the person performing the activity and its duration.

Produce a class diagram showing the following:

- a Crop Class with a Cropname and required FieldType attribute.
- A Field Class with attributes for FieldId and FieldType
- a Crop Management Event class, with
    - crop management event start and end times,
    - person performing the event,
    - comments
    - a method for calculating the duration of the event
- The FieldType is an enumerated field.

Total 10 marks

Marks

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Question 8

Compare and contrast Communication Diagrams and Sequence Diagrams. **10**

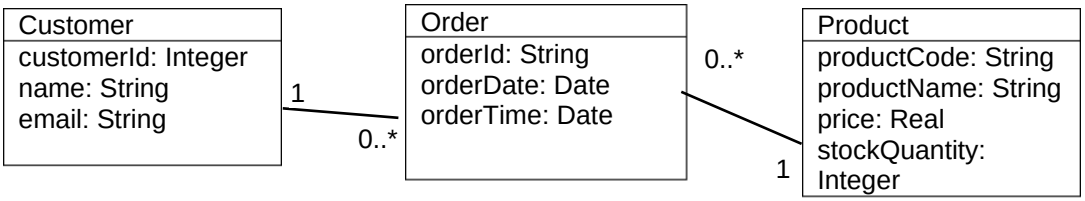
Total 10 marks

Marks

Question 9

a) Describe the purpose of using Object Constraint language (OCL) in Unified Modelling Language (UML). Identify **FOUR (4)** advantages it provides in terms of system specification **and** validation. **6**

b) Given the UML class diagram for a simple e-commerce application that includes the following classes: **4**



Using OCL, write constraints for the following requirements:

i) A Product must have a price greater than 0.

ii) A Customer must have a valid email address (must contain "@" and "."). The “@” must precede the “.”.

Total 10 marks

Marks

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Question 10

a) In object-oriented design, Explain the difference between Aggregation **and** Composition. **7**

Provide examples to illustrate your answer.

b) Provide an example of inheritance in the context of object-oriented design. **3**

Total 10 marks

END OF PAPER

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